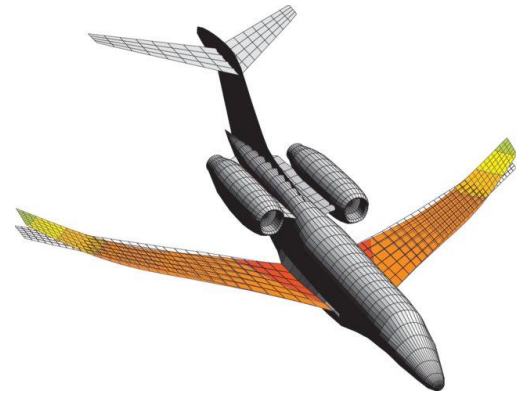


PROJECT TOPIC
Spring 2026

Vibration Absorber for a Generic Aircraft Wing

Ref Code : WINGVIBE
Type : New
Status : Group
Seats : 2 Student (Academic-Year Long)



Project Description:

The project involves vibration analysis and vibration absorber design for a given simple wing.

Work Description:

- Literature survey for vibration analysis of wings.
- Literature survey for frequency content (power spectral density) of atmospheric turbulence.
- Literature survey for frequency content of runway roughness for various airports.
- Literature survey for other sources of vibration for aircraft.
- Modeling the given wing with some complexity with a finite element code if possible.
- If finite element knowledge is lacking, the wing will be modeled in a simpler way using MATLAB.
- Vibration analysis to determine modal (natural) frequencies and mode shapes. (The wing here is a distributed mass and distributed stiffness system.)
- Designing internal and external absorber(s) for the wing (the latter is for academic curiosity; an external absorber would disturb the air flow over the wing).
- Vibration analysis with the absorber mounted to determine the reduction in vibration levels.

Required Qualifications and Skills:

- Appetite for theoretical/numerical analysis.
- Knowledge of the finite element method or MATLAB.
- The project will be on campus.

Related UN Sustainable Development Goals:

- No.9 Industry, innovation and infrastructure.

Difficulty:	high (X)	medium ()	low ()
Time commitment:	high (X)	medium ()	low ()
Estimated expense:	0 TL		